Isomorphic Questions

Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which of the following statements about simplifying assumptions in mechanics is most reasonable?

A. We can model a hummingbird's wing as a point mass when predicting the detailed bending motion during a wingbeat.
This confuses whole-body motion modeling with the need for an extended-body model to capture wing deformation.

B. We can model a city bus as a point mass when tracking its position change over several kilometers on a map.
Over kilometer-scale travel the bus's size is negligible compared with the distance, so a point-mass model is appropriate.

C. We are never allowed to treat the surroundings as a vacuum, even if we intend to neglect air drag in a calculation.
This is incorrect because "vacuum" is often used as an idealization to ignore air resistance when it is negligible.

D. None of the above statements is correct.
Option B is a valid modeling approximation, so "none" is not correct.

Q2. Which of the following equations is correct?

A. $1\ \text{mm} = 1\ \text{m}$
This ignores the metric prefix and incorrectly treats a millimeter as a meter.

B. $1 = 0.001$
This incorrectly equates two pure numbers with different values without units.

C. $1\ \text{mm} = 0.001\ \text{m}$
Since 1 m = 1000 mm, one millimeter equals 10^{-3} meters.

D. $1000\ \text{mm} = 0.001\ \text{m}$
This inverts the conversion factor because 1000 mm = 1 m, not 0.001 m.

Q3. Which option has the incorrect unit?

A. Power: watt
A watt is correctly the SI unit of power (J/s).

B. Momentum: kilogram· meter/second
Momentum correctly has units of kg m/s.

C. Pressure: newton/meter
Pressure should be force per area (N/m^2), so N/m is dimensionally wrong.

D. Electric potential: joule/coulomb
A volt is correctly J/C, so this unit matches electric potential.

Q4. A rescue drone is launched from a base at the origin and is located at $\vec{r} = 5\hat{\imath} - 2\hat{\jmath}$ (units: kilometers). Which of the following statements is correct?

A. The distance between the drone and the base is 7 km.
This adds component magnitudes instead of using the Pythagorean distance in 2D.

B. The distance between the drone and the base is 5 km.
This uses only the x-component and ignores the y-component contribution to distance.

C. The distance between the drone and the base is 2 km.
This uses only the y-component magnitude and ignores the x-component contribution to distance.

D. None of the above is correct.
The distance is $\sqrt{5^2 + (-2)^2} = \sqrt{29} \approx 5.4$ km, which is not 7, 5, or 2 km.

Q5. If the displacement of an object is given by the function $\vec{s} = \big((0.90\ \text{m/s}^3)t^3 + (1.60\ \text{m/s}^2)t^2\big)\hat{\imath}$, which of the following statements is correct?

A. This is a constant-acceleration motion.
Because $s(t)$ contains a t^3 term, the second derivative (acceleration) depends on t and is not constant.

B. The magnitude of the acceleration is 1.60 m/s^2.
This incorrectly reads off a coefficient from $s(t)$ instead of differentiating twice to get $a(t)$.

C. The magnitude of the acceleration at $t = 0$ is 0 m/s^2.
Differentiation gives a nonzero constant term in $a(t)$ at $t = 0$.

D. The magnitude of the acceleration at $t = 0$ is 3.2 m/s^2.
With $a(t) = d^2s/dt^2 = 6(0.90)t + 2(1.60)$, we get $a(0) = 3.2$ m/s^2.

Q6. An object takes 4 seconds to move from position A ($\vec{r}_A = 1.0\hat{\imath} - 3.0\hat{\jmath}$) to position B ($\vec{r}_B = -5.0\hat{\imath} + 4.0\hat{\jmath}$) along a straight path, where length units are in meters. Which of the following statements is correct?

A. The moving distance is 9.2 m.
For straight-line travel the distance equals $|\Delta\vec{r}| = \sqrt{(-6)^2 + 7^2} = \sqrt{85} \approx 9.2$ m.

B. The speed at $t = 0$ must be 2.3 m/s.
Knowing only endpoints and total time does not uniquely determine the initial speed without a motion model.

C. The acceleration at $t = 0$ must be 0.58 m/s^2.
Acceleration cannot be inferred from just the endpoints and elapsed time without additional information about the trajectory in time.

D. None of the above is correct.
Option A is correct because the straight-line distance is determined solely by the endpoints.

Q7. A soccer ball is kicked with a launch speed v_0 at an angle θ above the horizontal (neglect air resistance). What is the speed of the ball at its maximum height?

A. v_0
This ignores that the vertical component of velocity decreases to zero at the top.

B. $v_0 \cos\theta$
At maximum height the vertical velocity is zero, so the speed equals the unchanged horizontal component $v_0 \cos\theta$.

C. $v_0 \sin\theta$
This confuses the initial vertical component with the speed at the peak, where the vertical component is actually zero.

D. 0
Only the vertical component becomes zero at the top; the horizontal component remains nonzero.

Q8. A probe is traveling with an initial velocity of $\vec{v}_0 = (15.0\ \text{m/s})\hat{\imath}$ when its thrusters fire, imparting an acceleration of $\vec{a} = (-1.2\ \text{m/s}^2)\hat{\imath} + (3.4\ \text{m/s}^2)\hat{\jmath}$. Based on this scenario, which of the following statements is correct?

A. The x-component of the probe's velocity remains constant.
Because $a_x \neq 0$, the x-component of velocity must change with time.

B. The y-component of the probe's velocity remains constant.
Because $a_y \neq 0$, the y-component of velocity must change with time.

C. Both the direction of motion and speed change.
Nonzero acceleration components in both x and y alter the velocity vector's magnitude and its direction over time.

D. None of the above is correct.
Option C correctly describes the consequence of having both a_x and a_y nonzero.

Q9. If the magnitudes of both charges are multiplied by 4 and their separation is doubled, what happens to the magnitude of the electrostatic force that each charge exerts on the other?

A. It stays the same.
This would require the numerator and denominator scaling to cancel, which they do not here.

B. It decreases by a factor of 4.
This incorrectly treats increasing charges as reducing the force or overemphasizes the distance change.

C. It increases by a factor of 2.
This results from applying a linear (not squared) dependence on separation or mishandling the scaling factors.

D. It increases by a factor of 16.
This ignores the r^2 in Coulomb's law and accounts only for the charge scaling.

E. It increases by a factor of 4.
By Coulomb's law $F \propto q_1 q_2 / r^2$, so the factor is $(4 \cdot 4)/(2^2) = 16/4 = 4$.

Q10. An object can escape a planet with radius R and mass M at an escape speed given by $v = \sqrt{\frac{2GM}{R}}$, where G is the universal gravitational constant. Based on this expression for escape speed, which of the following statements is correct?

A. A more massive spacecraft needs a larger escape speed from the same planet.
The spacecraft's mass does not appear in $v = \sqrt{2GM/R}$, so escape speed is independent of the object's mass.

B. If a planet's radius increases while its mass stays the same, the escape speed increases.
Since $v \propto 1/\sqrt{R}$, increasing R actually decreases the escape speed when M is fixed.

C. A planet with smaller mass and larger radius requires a higher escape speed.
Both smaller M and larger R reduce M/R, so they lower the escape speed rather than raise it.

D. A planet with greater mass and smaller radius requires a higher escape speed.
Because $v \propto \sqrt{M/R}$, increasing M and decreasing R both increase M/R and thus increase v.

Q11. The force $\vec{F} = (2.50\ \text{N})\hat{\imath} + (6.00\ \text{N})\hat{\jmath}$ acts on a particle as it moves along the x axis from $x = -1.2$ m to $x = 3.8$ m. What is the work, in joules, done by the force on the particle?

A. 12.5 J
Because the displacement is along x, $W = \vec{F} \cdot \Delta\vec{r} = F_x \Delta x = 2.50 \times (3.8 - (-1.2)) = 12.5$ J.

B. 30.0 J
This incorrectly uses only the y-component of force even though there is no y-displacement.

C. 42.5 J
This mistakenly adds components (or magnitudes) instead of taking the dot product with the displacement direction.

D. 7.5 J
This comes from using an incorrect displacement (e.g., forgetting that Δx spans from -1.2 m to 3.8 m).

Q12. A 3.0 kg cart initially moving at 4.0 m/s collides with a bumper and rebounds at a speed of 1.5 m/s in the opposite direction. The collision lasts for 0.10 seconds. What is the average force exerted on the cart during the collision (magnitude)?

A. 75 N
This corresponds to using only $mv_i/\Delta t$ and ignores the rebound (the final velocity).

B. 135 N
This results from using $\Delta v = 4.5$ m/s (adding speeds but not accounting correctly for opposite directions).

C. 165 N
Using impulse, $|F_{\text{avg}}| = m|\Delta v|/\Delta t = 3.0\,|(-1.5 - 4.0)|/0.10 = 165$ N.

D. None of the above is correct.
Option C matches the impulse/average-force calculation, so "none" is not correct.

Q13. A 70 kg player is initially at rest on ice. A 0.20 kg ball moving at 120 km/hr is caught and held by the player (they move together afterward). What is the player's speed immediately after the catch?

A. 0.067 m/s
This typically comes from dividing by 70 kg instead of the combined mass (70.20 kg) or using an incomplete conversion.

B. 0.095 m/s
Conservation of momentum gives $v = \frac{m_b v_b}{m_p + m_b} = \frac{0.20\,(120\ \text{km/hr} = 33.3\ \text{m/s})}{70.2} \approx 0.095$ m/s.

C. 33.3 m/s
This ignores the large mass of the player and incorrectly assumes the final speed equals the ball's speed.

D. 9.5 m/s
This usually results from a major unit-conversion mistake (e.g., treating km/hr as m/s or missing a factor of 3.6).
